

## Highlights

### **Designing Multi-Compartment Last-Mile Vehicle Fleets: An Open-Source Matheuristic**

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- We study a fleet size and mix problem with heterogeneous multi-compartment vehicles.
- We propose a novel matheuristic approach based on a cluster-first, fleet design-second scheme.
- We present FLEETMIX, a spec-driven, open-source implementation of our matheuristic.
- We show the effectiveness of FLEETMIX compared to existing solution approaches.
- We derive several managerial insights for a real-world case study.

# Designing Multi-Compartment Last-Mile Vehicle Fleets: An Open-Source Matheuristic

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## Abstract

Urban food distributors are increasingly relying on flexible, multi-compartment vehicle fleets to handle diverse product mixes and challenging urban environments. While recent research on multi-compartment vehicle routing problems has explored a range of heuristics and constraints, to our knowledge, none has directly addressed fleet size and mix optimization for heterogeneous vehicles with flexible compartments in food distribution systems. Motivated by our collaboration with a major food distributor operating in Colombia, we close this practical and methodological gap by proposing a novel cluster-first, fleet design-second matheuristic approach that quickly produces cost-effective fleet size and mix solutions. To support practitioners and researchers in applying and replicating our results, we make our matheuristic publicly available through FLEETMIX, an open-source Python package freely available under the MIT license and distributed via PyPI. We show the effectiveness of FLEETMIX in finding high-quality solutions compared to existing solution approaches on available benchmark instances. Based on a real-world case study from our industry partner, which serves hundreds of customers daily, we also show the value of utilizing multi-compartment vehicles to reduce fleet sizes and total logistics costs.

*Keywords:* fleet design, multi-compartment, matheuristic, open-source Python package, last mile

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## 1. Introduction

The growing expansion of the food industry has drawn increasing attention to food safety and quality management during production, storage, and transportation (Hsiao et al., 2017). Food service companies and grocery stores typically receive a range of product categories from distributors, each requiring specific temperature conditions defined by legal regulations or quality standards (Hertog et al., 2014; Ostermeier and Hübner, 2018). Consequently, food distributors must be able to transport multiple temperature-sensitive product types simultaneously. However, due to their

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distinct environmental needs, these products cannot be stored together in the same space during transit (Chen et al., 2019). This makes planning efficient delivery operations both increasingly challenging and essential, as delivery costs account for 20% of overall food supply chain costs (Kuhn and Sternbeck, 2013).

To address this challenge, distributors have increasingly turned to fleets of Multi-Compartment Vehicles (MCVs) in recent years, which enable the simultaneous transport of multiple product categories (Ostermeier et al., 2021). MCVs are generally designed with either fixed or flexible compartments. MCVs with fixed compartments have predefined capacities determined during manufacturing and cannot be altered, whereas MCVs with flexible compartments offer the ability to adjust their sizes (Henke et al., 2015). In the context of food distribution, MCVs commonly have three compartments for dry, refrigerated, and frozen foods, with adjustable sizes to match load requirements. Companies like FreshDirect, Sysco, and Whole Foods Markets have reported the benefits of using MCVs, including improved customer experience through single-stop deliveries and reduced fuel costs, emissions, traffic exposure, and parking issues (He et al., 2024). By consolidating different product categories, MCVs also reduce the number of vehicles needed to meet demand compared to using separate Single-Compartment Vehicles (SCVs).

Motivated by our collaboration with a major food distributor operating in Bogotá, Colombia, we investigate the problem of designing MCV fleets for last-mile food distribution systems. Unlike existing works, our problem considers several important real-world features of modern delivery operations with MCVs, including heterogeneous vehicles with flexible compartments, both fixed and variable vehicle usage costs, route duration constraints, and a large number of customers. The problem consists of determining three interconnected decisions: (i) the number of each vehicle type to use, (ii) the type, number, and size of compartments assigned to each vehicle, and (iii) the distribution plan (i.e., demand allocation) for each vehicle. The objective is to minimize total fixed and variable vehicle costs while satisfying customer demand for different product types.

Our main methodological and practical contributions can be summarized as follows.

1. **Novel Matheuristic Approach.** We propose a matheuristic approach based on a cluster-first, fleet design-second scheme. In this approach, we first generate feasible, potentially overlapping customer clusters using tailored clustering techniques that group customers while satisfying both vehicle capacity and maximum route duration constraints. We then use a mathematical programming formulation to determine the optimal MCV fleet size and mix that minimizes total costs while satisfying each individual customer’s demand. An appealing property of our matheuristic is that it can be easily adapted to existing variants addressed in the literature (see Section 2), including homogeneous vehicles, fixed compartments, and single- and multi-stop policies (where customers are allowed to be served via a single or multiple vehicle stops, respectively). This makes our matheuristic a unified and versatile approach suitable for a wide range of applications.
2. **Open-Source Python Package.** To support practitioners and decision-makers in apply-

ing and replicating our findings, we introduce FLEETMIX, an open-source Python package available on GitHub (<https://github.com/ekohan/fleetmix>) under the MIT license and distributed via PyPI. FLEETMIX features a modular, protocol-based architecture that allows researchers to replace individual algorithmic components without compromising the integrity of the overall matheuristic pipeline. To further facilitate adoption, we provide three user interfaces: a command-line interface, a Python API, and a web-based graphical interface tailored for fleet managers with no coding experience.

3. **Numerical Validation and Managerial Insights.** We test our matheuristic approach using both synthetic instances from the literature and a real-world dataset from a food distributor operating in Bogotá, Colombia, which serves hundreds of customers daily. Our results show the effectiveness of our matheuristic approach compared to benchmark solution approaches, generating high-quality fleet designs in seconds. For real-world instances, we illustrate the benefits of using MCVs to increase vehicle utilization and reduce both total costs and the number of vehicles compared to traditional SCV fleets.

The rest of the manuscript is structured as follows. We review the relevant literature in Section 2, highlighting the research gaps we address and the main contributions of our work. Section 3 defines the problem under investigation. In Section 4, we describe our cluster-first, fleet design-second solution approach. Section 6 shows the effectiveness of our matheuristic approach over synthetic instances from the literature, while Section 7 presents the results obtained using the real-world case study. We then summarize the main managerial implications of our work in Section 8. Lastly, Section 9 outlines our conclusions and describe several research avenues for future research.

## 2. Literature Review

In this section, we review three fields of work related to MCVs. We begin by briefly analyzing the literature related to Vehicle Routing Problems (VRPs). We then concentrate on VRPs with MCVs, and finally, we review the literature related to fleet composition problems. We conclude this section by describing the main research gaps and contributions of our work.

### 2.1. Vehicle Routing Problems

VRP is a long-lasting topic that has been researched for over 60 years (Laporte, 2009). The canonical VRP aims to find routes from a depot to geographically scattered customers at minimal cost. Its basic assumptions are that vehicles are homogeneous, there is a single depot, customer demand is completely known at the time of delivery planning, and there is a symmetric cost matrix (Laporte, 2009).

The VRP is mathematically classified as an NP-hard problem, meaning that optimal solutions are often very hard to achieve, especially when considering large instance sizes. To address this

challenge, most literature has focused on developing heuristic approaches that can yield near-optimal solutions with much less computational power and time (see, e.g., [Prins et al. \(2014\)](#) and [Braekers et al. \(2016\)](#)). A popular heuristic framework is the so-called cluster-first, route-second approach, where demand/customer locations are first grouped while respecting the capacity constraints. Then, a route is determined separately for each cluster by solving a Traveling Salesman Problem (TSP) ([Laporte, 2009](#)). The most popular examples are the sweep heuristics proposed by [Gillett and Miller \(1974\)](#) and [Fisher and Jaikumar \(1981\)](#), where the customers served by each route are determined according to the polar coordinate angle of each location. We refer the reader to [Miranda-Bront et al. \(2017\)](#), [Comert et al. \(2018\)](#), and [Alesiani et al. \(2022\)](#) for recent works proposing cluster-first, route-second heuristics.

Current VRP models have evolved to incorporate practical challenges such as traffic influences on time, time windows, and dynamic demand ([Braekers et al., 2016](#)). There are several variants of the VRP that have been extensively discussed in the literature, such as the capacitated VRP, which considers vehicle capacity constraints; the VRP with time windows, where customers must be visited within specific time windows; dynamic VRPs, where customer information is revealed over time; and multi-depot VRPs, among others ([Braekers et al., 2016](#)). The interested reader can refer to [Vidal et al. \(2020\)](#) for a concise review of existing and emerging VRP variants.

## 2.2. Multi-Compartment Vehicle Routing Problems

The Multi-Compartment Vehicle Routing Problem (MCVRP) is one of the most important VRP variants that considers that vehicles' load capacity can be divided into different compartments. A recent comprehensive review of MCVRPs is presented by [Ostermeier et al. \(2021\)](#), who propose a taxonomy based on attributes related to compartment characteristics and order fulfillment. As highlighted by [Ostermeier et al. \(2021\)](#), MCVRPs find many practical applications, including fuel distribution, waste collection, agricultural supply chains, maritime transportation, and food distribution. However, most literature focused on MCVs with fixed compartments, meaning that the number and capacity of compartments cannot be adjusted to match load requirements. For instance, [Mendoza et al. \(2010\)](#) address a MCVRP with stochastic demands, assuming fixed compartment sizes and an unlimited fleet of homogeneous vehicles. The authors formulate the problem as a two-stage stochastic programming model and develop a memetic algorithm with solution evaluation and reparation strategies that account for the stochastic nature of the problem. [Mendoza et al. \(2010\)](#) show the effectiveness of their memetic algorithm using synthetic instances with up to 480 customers. Later, [Reed et al. \(2014\)](#) study a MCVRP in the context of household solid waste recycling collection. They consider a homogeneous fleet with two fixed compartments and propose an ant colony heuristic approach. The authors test their approach on synthetic instances with 50 customers and two types of products.

To the best of our knowledge, [Henke et al. \(2015\)](#) are the first to introduce the MCVRP with flexible compartment sizes in the context of glass waste collection. Assuming a predefined, homogeneous fleet of vehicles (i.e., no fleet sizing decisions are considered), the authors formulate a

Mixed-Integer Linear Programming (MILP) model to minimize the total distance traveled by the vehicles. [Henke et al. \(2015\)](#) also devise a variable neighborhood search algorithm that allows visiting infeasible solutions in terms of vehicle capacity. Their experiments show that their MILP model can solve instances with up to 10 customers and apply their heuristic to instances with up to 50 customers. Later, [Henke et al. \(2019\)](#) propose a branch-and-cut algorithm to address the problem they had previously introduced in [Henke et al. \(2015\)](#). We refer the reader to [Ostermeier et al. \(2021\)](#) for other related works focusing on routing MCVs.

Now, in the context of food distribution, [Hübner and Ostermeier \(2019\)](#) investigate the impact of loading and unloading costs when routing a homogeneous fleet of MCVs with flexible compartments. The authors propose a large neighborhood search heuristic with tailored removal and insertion operations. Their experiments show that MCVs can reduce total costs by up to 6.3% compared to a SCV fleet. Another study that considers two-dimensional loading constraints can be found in [Ostermeier et al. \(2018\)](#). Next, [Chen et al. \(2019\)](#) study a MCVRP with time windows and propose a Adaptive Large Neighborhood Search (ALNS) heuristic. Their heuristic is applied to a real-world case study from a cold-chain distribution company in Shanghai, showing that total costs increase as time windows become tighter, mainly due to higher fuel consumption. Concurrently, [Martins et al. \(2019\)](#) consider a MCVRP with product-oriented time-window constraints. The authors also propose an ALNS heuristic, which is tested using instances from [Kovacs et al. \(2014\)](#). [Martins et al. \(2019\)](#) show that scheduling time windows in advance, rather than assigning them after the daily plans are made, leads to more on-time deliveries. Later, [Beneich and Douiri \(2023\)](#) develop a hybridized Simulated Annealing (SA) algorithm. Their SA algorithm is applied to 28 problems, yielding competitive results compared to the heuristics proposed by [Abdulkader et al. \(2015\)](#) and [Yahyaoui et al. \(2020\)](#). The interested reader can refer to the recent literature review presented by [Guerriero et al. \(2025\)](#) for other works addressing VRPs with perishable products.

### *2.3. Fleet Size and Mix Problems*

As can be observed in the previous section, most literature on MCVs focuses on routing a predefined and fixed fleet of homogeneous vehicles, without considering decisions about fleet size and mix. While fleet size and mix problems have recently attracted growing attention due to their practical relevance, most literature concentrates on distribution problems involving SCVs ([Hoff et al., 2010](#)). Specifically, since the seminal work of [Golden et al. \(1984\)](#), who introduce the fleet size and mix problem considering SCVs with fixed acquisition and variable routing costs, the literature has progressively incorporated various real-world features, such as delivery time windows ([Liu and Shen, 1999](#); [Bräysy et al., 2008](#); [Repoussis and Tarantilis, 2010](#)), limited number of vehicles ([Koç et al., 2016](#)), electric vehicles ([Hiermann et al., 2016](#); [González-Rodríguez et al., 2024](#)), and stochastic customer demand ([Loxton et al., 2012](#); [Zinnenlauf et al., 2024](#)), among others. However, to the best of our knowledge, the only contribution addressing fleet size and mix decisions with MCVs has been proposed by [Ostermeier and Hübner \(2018\)](#).

Ostermeier and Hübner (2018) examine a MCVRP with vehicle selection decisions. However, the authors only consider two homogeneous vehicle types: SCVs and MCVs, meaning that all vehicles within each type have identical characteristics (e.g., the same maximum load capacity). Ostermeier and Hübner (2018) propose a mathematical programming formulation aiming to find the number and routes for SCVs and MCVs that minimizes a non-linear cost function composed by fixed cost per active compartment, transportation costs based on distance traveled, and a fixed cost per vehicle stop. They also propose a ALNS heuristic based on their previous work in Hübner and Ostermeier (2019). Using a real-world case study from a grocery retailer, Ostermeier and Hübner (2018) show that using a mixed fleet of SCVs and MCVs can reduce total costs by up to 30%.

#### 2.4. Research Gaps and Contributions

Our literature review identifies several practical and methodological research gaps. Most existing studies have focused on routing MCVs with fixed compartments, implying that vehicle capacities cannot be adapted to better match customer demand (Ostermeier and Hübner, 2018). While some studies address MCVRP with flexible compartments, such as Henke et al. (2015) and Hübner and Ostermeier (2019), their scope is limited to operational decisions related to routing and demand loading, without considering decisions on fleet size and mix. The work most closely related to ours is that of Ostermeier and Hübner (2018). Yet, Ostermeier and Hübner (2018) do not consider important real-world features of food delivery operations, including heterogeneous vehicles and maximum route duration constraints. Consequently, we propose the following contributions to the academic literature.

First, we investigate a comprehensive variant of the fleet size and mix problem arising in modern food distribution operations. As we describe in further detail in Section 3, our tactical decision-making problem includes heterogeneous MCVs, flexible compartments, fixed costs per vehicle and compartment used, variable transportation costs, single- and multi-stop delivery policies, and maximum vehicle capacity and route duration constraints.

Second, given our interest in addressing practical food distribution operations involving hundreds of customers, we propose a novel matheuristic approach based on a cluster-first, fleet-design second scheme. Our matheuristic is based on our newly introduced concept of *vehicle configuration*, which facilitates modeling heterogeneous fleets with flexible compartments by capturing the different ways each vehicle type can be compartmentalized. Leveraging this concept, our matheuristic creates feasible customer clusters per vehicle configuration, satisfying both capacity and route duration constraints. An integer programming model then selects the optimal number of vehicles and their compartmentalization (i.e., the number, type, and size of compartments) to serve each individual customer’s demand. The fleet size and mix found is then enhanced through an improvement phase that iteratively includes additional customer clusters into our integer program.

Third, we develop FLEETMIX, an open-source Python package that allows practitioners and researchers to apply our matheuristic and replicate our results. FLEETMIX features a modular,

protocol-based architecture that allows replacing individual components while preserving the overall methodological flow. To accommodate a wide range of users, FLEETMIX includes three interaction modes: a command-line tool, a Python API, and an intuitive web-based interface developed specifically for fleet managers without coding expertise.

Lastly, we validate the effectiveness of our matheuristic approach using both synthetic instances from the literature and a case study informed by real-world data from a food distributor in Bogotá, Colombia. [Include a sentence summarizing the main results over synthetic instances]. Based on our real-world case study, we derive important managerial insights regarding the effects of [to be completed].

### 3. Problem Definition

We now formally introduce the problem under investigation. We consider a single depot with sufficient inventory at the beginning of the delivery horizon to satisfy all customer demands. Let  $N' = \{0, 1, \dots, n\}$  be the set of nodes, where 0 denotes the depot and  $N = \{1, \dots, n\}$  the customers. Further, let  $P$  be the set of product types (e.g., dry, chilled, and frozen) and  $q_p$  the units of vehicle load capacity required to transport one unit of product type  $p \in P$ . Each customer  $i \in N$  has a demand of  $d_{ip}$  units of product type  $p \in P$ . The food distributor that inspired this work employs a *single-stop policy*, which dictates that the demand for each product type from each customer must be fulfilled by a single vehicle. Nevertheless, we also examine a multi-stop policy in which customers may receive deliveries from multiple vehicles, but each product type must still be delivered in full by one vehicle, as splitting is not practical (Gu et al., 2024).

The delivery fleet comprises a set of heterogeneous vehicle types, each defined by specific operational characteristics such as maximum load capacity, average speed, and fixed and variable costs. Each vehicle type can divide its load capacity into a set  $M$  of flexible compartments, each designed for a specific product type. In other words, vehicles can be configured in multiple ways by assigning different subsets of compartments. To capture this flexibility, we introduce the concept of *vehicle configuration*, defined as a certain vehicle type equipped with a (non-empty) subset of compartments. The set of all vehicle configurations is denoted by  $V$ . For example, if the company operates two vehicle types and there are three available compartments, each type can be configured in  $2^3 - 1 = 7$  distinct ways, resulting in  $2 \times 7 = 14$  configurations in  $V$ . Based on the corresponding vehicle type, each vehicle configuration  $v \in V$  has a maximum load capacity of  $Q_v$ , which can be distributed among its compartments, and a maximum route time duration of  $T_v$ .

Given our interest in addressing practical food distribution problems involving hundreds of customers, we aggregate customers into multiple capacitated clusters, denoted by the set  $K$ . Since these clusters are not necessarily mutually exclusive subsets of  $N$ , we define  $K_i \subseteq K$  as the subset of clusters containing customer  $i \in N$ . As we explain in further detail in Section 4, clusters are created based on the set of vehicle configurations  $V$ . Notably, we define  $K_v \subseteq K$  as the subset of clusters that can be feasibly served by vehicle configuration  $v \in V$  regarding maximum load capacity ( $Q_v$ )



and route time duration ( $T_v$ ) constraints. Similarly, let  $V_k \subseteq V$  be the set of vehicle configurations that can feasibly serve cluster  $k \in K$ . Based on the business context considered in our work, we consider that each vehicle used performs a single trip during the delivery horizon (e.g., a day). The total cost of serving cluster  $k \in K_v$  with vehicle configuration  $v \in V$ , denoted by  $c_{vk}$ , consists of a fixed cost incurring for dispatching the vehicle and equipping it with the set of compartments associated with the configuration, along with a variable cost based on the distance traveled to serve all customers in the cluster. Table [TBD](#) in [Appendix TBD](#) summarizes our mathematical notation.

The problem involves determining the optimal MCV fleet size and mix by deciding the number of each vehicle configuration to use. Specifically, this includes selecting the appropriate vehicle types, defining their compartmentalization (i.e., number, type, and load capacity associated with each compartment), and assigning each selected vehicle configuration to a specific customer cluster. The objective is to minimize the total fixed and variable vehicle costs while meeting the individual demand of each customer for different product types. Note that, as we address a tactical fleet size and mix problem, operational routing decisions are not explicitly modeled but are implicitly accounted for when verifying that each customer cluster adheres to maximum route time constraints associated with the respective vehicle configuration.

#### 4. Matheuristic Approach

Given our interest in designing MCV fleets for practical food distribution operations, we propose a novel matheuristic approach based on a cluster-first, fleet design-second scheme (see [Archetti and Speranza \(2014\)](#) for a survey on matheuristics applied to VRPs). [Figure 1](#) presents a high-level overview of the main steps involved. First, we define the set of possible vehicle configurations based on the available vehicle types and compartments. Subsequently, for each vehicle configuration, we generate a set of feasible customer clusters using tailored clustering techniques. We then determine the optimal fleet size and mix through an integer programming model. To further enhance the solution, we apply an improvement phase to create and include additional feasible customer clusters. This procedure repeats until no further improvements can be reached, at which point we return the best fleet size and mix found.

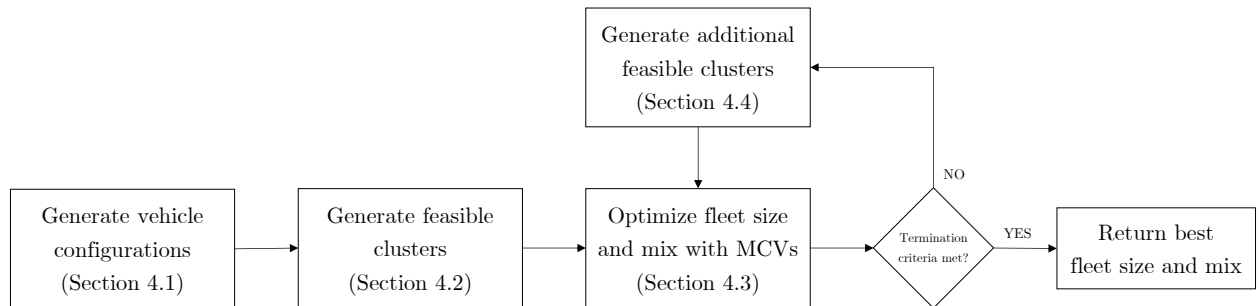


Figure 1: High-level overview of the matheuristic approach.

#### 4.1. Generate Vehicle Configurations

We define the set of vehicle configurations,  $V$ , based on the available vehicle types and compartments. Specifically, for each vehicle type, we define  $2^{|M|} - 1$  configurations, where  $M$  denotes the set of compartments. For example, the food distributor that inspired this research divides their vehicles into three compartments dedicated to dry ( $m^{\text{dry}}$ ), refrigerated ( $m^{\text{ref}}$ ), and frozen ( $m^{\text{fro}}$ ) product types. In this case, we consider the following subsets of  $M$ , each defining a configuration:  $\{m^{\text{dry}}\}$ ,  $\{m^{\text{ref}}\}$ ,  $\{m^{\text{fro}}\}$ ,  $\{m^{\text{dry}}, m^{\text{ref}}\}$ ,  $\{m^{\text{dry}}, m^{\text{fro}}\}$ ,  $\{m^{\text{ref}}, m^{\text{fro}}\}$ , and  $\{m^{\text{dry}}, m^{\text{ref}}, m^{\text{fro}}\}$ . Therefore, we generate  $r \cdot (2^{|M|} - 1)$  vehicle configurations, with  $r$  denoting the number of vehicle types.

Note that our procedure includes generating SCVs dedicated to a particular product type (namely, the configurations arising from the singletons  $\{m^{\text{dry}}\}$ ,  $\{m^{\text{ref}}\}$ , and  $\{m^{\text{fro}}\}$ ), enabling our fleet size and mix optimization model to select either MCVs or SCVs based on their respective cost structures and characteristics. Since we consider multiple heterogeneous vehicle types, our procedure generalizes the approach presented by [Ostermeier and Hübner \(2018\)](#), who only consider one vehicle type. Lastly, since we consider MCVs with flexible compartments, note that our optimization model then defines the specific load capacity associated with each compartment of the selected vehicle configurations.

#### 4.2. Generate Feasible Customer Clusters

After defining the set of vehicle configurations,  $V$ , we generate feasible clusters of customers that can be served by at least one configuration while satisfying maximum load capacity and route duration constraints. Specifically, for each configuration  $v \in V$ , we apply a variety of clustering techniques to create clusters that are feasible for that configuration. These techniques include  $k$ -means,  $k$ -medoids, Gaussian mixture models, and hierarchical (agglomerative) clustering (see [Appendix A](#) for a summary of the clustering techniques). While these are standard clustering techniques, we adapt them to account for both geographical and demand proximity between customers. Furthermore, we apply a split-and-merge procedure to the resulting set of clusters, not only to validate feasibility but also to enrich the final set of clusters that serves as input to our optimization phase.

Note that, as described in [Section 3](#), the resulting clusters do not necessarily partition the set of customers, as some customers may appear in more than one cluster. However, the actual allocation of each customer to a given cluster (and, therefore, vehicle configuration) is then determined during our optimization phase (see [Section 4.3](#)).

In the following, we explain how load capacity and route duration feasibility are verified, as well as how the distance metric is adapted for the clustering techniques used.

##### 4.2.1. Load Capacity and Route Duration Feasibility Checks

To ensure the creation of feasible customer clusters, we perform both capacity and route duration feasibility checks. Capacity constraints require that the total demand across all product types within a cluster does not exceed the total capacity of the corresponding vehicle configuration, while route

duration constraints ensure that the total travel time to serve a cluster remains within a prescribed threshold. Since the total capacity required by a cluster can be readily determined by summing the demand across all product types, we now focus on how route durations are computed within the context of our tactical decision-making problem. We then explain how both capacity and route duration feasibility are ensured through a split-and-merge procedure.

*Computation of vehicle route durations.* Depending on the application context, our matheuristic allows decision-makers to verify maximum route duration constraints using one of two methods: Continuous Approximation (CA) or a state-of-the-art routing solver. CA is a widely used technique to approximate vehicle routing costs in large-scale integrated decision-making problems where operational routing decisions play a subordinate role, such as location-routing (Winkenbach et al., 2016; Pina-Pardo et al., 2022), workforce sizing and routing (Mandal et al., 2025), and fleet design and routing problems (Franceschetti et al., 2017; Zinnenlauf et al., 2024). In this work, we consider the seminal CA formula proposed by Beardwood et al. (1959), extended to incorporate vehicle setup times, customer service times, and depot-to-cluster travel times. Specifically, for a given cluster  $k \in K_v$  with  $n$  customers distributed over a service area  $A$ , we approximate the total routing time of serving the cluster using vehicle configuration  $v \in V$  as

$$t_{vk} \approx \alpha_{vk} + 2 \cdot \delta_{vk} + \beta \cdot \sqrt{n \cdot A} + \gamma \cdot n, \quad (1)$$

where  $\alpha_{vk}$  is the setup time to dispatch the vehicle configuration,  $\delta_{vk}$  the line-haul travel time between the depot and the centroid of the cluster,  $\gamma$  the customer service time, and  $\beta$  is a non-negative constant. The third term corresponds to the well-known Bearwood-Halton-Hammersley (BHH) approximation presented by Beardwood et al. (1959) (see Silva et al. (2022) for a recent work showing the quality of this approximation to produce near-optimal estimates).

Note that an alternative method for computing vehicle route durations is to employ an existing routing solver. In our matheuristic implementation (see our GitHub repository), decision-makers can also use the PYVRP library developed by Wouda et al. (2024), which is recognized as a state-of-the-art vehicle routing solver. While this approach might enhance vehicle routing time estimations, it often requires substantially more computational time than the CA-based approach, especially when dealing with large instances.

*Split-and-merge procedure.* After computing vehicle route durations, we verify both load capacity and route duration constraints through a split-and-merge procedure. First, we conduct a recursive binary splitting, as detailed in Algorithm 1. When cluster  $k \in K_v$  exceeds either the total capacity or the maximum route duration limit associated with vehicle configuration  $v \in V$  (i.e., when  $\sum_{i \in k} \sum_{p \in P} d_{ip} > Q_v$  or  $t_{vk} > T_v$ , respectively), the cluster is recursively divided into two subclusters. This splitting process continues until all resulting clusters satisfy both constraints or a predefined maximum recursion depth is reached.

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**Algorithm 1** Recursive Cluster Splitting

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```
1: function SPLITCLUSTER(Cluster, DepthLevel, MaxDepth)
2:   if EXCEEDSCONSTRAINTS(Cluster) and DepthLevel < MaxDepth then
3:     SubClusters  $\leftarrow$  SPLIT(Cluster, 2) ▷ Split cluster into two sub-clusters
4:     for each SubCluster in SubClusters do
5:       SPLITCLUSTER(SubCluster, DepthLevel + 1, MaxDepth)
6:     end for
7:   else
8:     ADDVALIDCLUSTER(Cluster)
9:   end if
10: end function
```

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Since Algorithm 1 may produce small clusters with only a few customers, we then apply a merging step to all clusters  $k \in K$  such that  $|k| < \eta$ , where  $\eta$  denotes a minimum cluster size. For each such small cluster  $k_i$ , we define a neighborhood consisting of **all other small clusters within a centroid-to-centroid travel distance of  $\Delta$** . Then, for each neighboring cluster  $k_j$ , we check whether the combined cluster  $k_i \cup k_j$  can be feasibly served by a single vehicle configuration, considering both load capacity and route duration constraints. If feasible, we update the set of clusters by adding  $k_i \cup k_j$ , i.e.,  $K = K \cup \{k_i \cup k_j\}$ . Note that a similar merging procedure is used during our improvement phase, but only to promising clusters selected in optimal solutions (see Section 4.4).

#### 4.2.2. Distance Metric

Lastly, we describe how we modify the distance metric to account for both geographic and demand proximity when generating clusters. To enhance the clustering process, we create a composite distance metric that combines geographical distance and product demand similarity into a single value. We thus define a composite distance function  $D_{ij}$  between customers  $i, j \in N$  as

$$D_{ij} = \lambda \cdot D_{ij}^{\text{geo}} + (1 - \lambda) \cdot D_{ij}^{\text{prod}} \quad (2)$$

where  $\lambda \in [0, 1]$  establishes the weight between geographical proximity, measured by the geographical distance  $D_{ij}^{\text{geo}}$ , and product demand similarity, given by the product distance  $D_{ij}^{\text{prod}}$  (we refer the reader to [Appendix A](#) for the specific value of  $\lambda$  used in each clustering technique considered). Notably, we compute the product distance (measuring demand similarity) as the cosine distance between weighted **demand composition profiles, where demands are normalized by total volume and weighted by product-specific temperature sensitivity**.

#### 4.3. Optimize Fleet Size and Mix with Heterogeneous MCVs

Based on the generated set of clusters, we now define the MCV fleet size and mix that minimizes the total costs incurred to meet each individual customer's demand. It is worth recalling that the

total cost of serving cluster  $k \in K_v$  with vehicle configuration  $v \in V$ , denoted by  $c_{vk}$ , includes both a fixed cost (associated with setting up the vehicle's compartments and dispatching it) and a variable cost that depends on the distance traveled to serve the cluster. Notably, such fixed costs capture the additional operational effort required to set up and operate a MCV configuration, as opposed to a SCV configuration.

Let  $x_{vk}$  be a binary decision variable equal to 1 if and only if vehicle configuration  $v \in V$  serves cluster  $k \in K_v$ . Our integer programming model reads

$$\text{minimize} \quad \sum_{v \in V} \sum_{k \in K_v} c_{vk} \cdot x_{vk}, \quad (3a)$$

subject to

$$\sum_{k \in K_i} \sum_{v \in V_k} x_{vk} = 1, \quad i \in N, \quad (3b)$$

$$\sum_{v \in V_k} x_{vk} \leq 1, \quad k \in K, \quad (3c)$$

$$x_{vk} \in \{0, 1\}, \quad v \in V, k \in K_v. \quad (3d)$$

The objective function (3a) minimizes the total fixed and variable costs. Constraints (3b) force to serve each individual customer  $i \in N$ . Constraints (3c) ensure that each cluster is visited by at most one vehicle configuration. Lastly, Constraints (3d) define the domain of the decision variables.

Note that, after solving Problem (3), the specific load capacity required for each compartment of the selected vehicle configurations can be readily determined by summing the demand for each product type across all customers in the respective cluster.

*Multi-stop delivery policy.* In line with the business context that motivates our work, Constraints (3b) enforce a single-stop policy, where the entire demand for each product type of each customer must be fully served by a single vehicle. However, in other contexts, it is acceptable for customers to receive multiple deliveries. In such cases, while multiple vehicle stops are allowed, splitting the delivery of a single product type is not impractical (Henke et al., 2015; Gu et al., 2024).

Let us introduce the following additional notation to accommodate this multi-stop delivery policy. Let  $P_i$  be the set of product types demanded by customer  $i \in N$ . We define  $S_i^p = \{P' \subseteq P_i : p \in P'\}$  as the set of all subsets of  $P_i$  that contain product  $p \in P_i$ . Lastly, for each product type  $p \in P_i$ , let  $K_i^s$  be the subset of clusters containing customer  $i \in N$  that can serve the demand of all product types in  $s \in S_i^p$ . Thus, the following constraints

$$\sum_{s \in S_i^p} \sum_{k \in K_i^s} \sum_{v \in V_k} x_{vk} = 1, \quad i \in N, p \in P_i, \quad (3e)$$

must replace Constraints (3b) in Problem (3) to allow multiple deliveries per customer.

#### 4.4. Improvement Phase

We perform an improvement phase on the MCV fleet size and mix found by Problem (3). Specifically, after solving Problem (3) for a given set of clusters  $K$ , we consider each pair of clusters  $(k_i, k_j)$  selected in the optimal solution  $\mathbf{x}^*$  (i.e.,  $x_{k_i, v_i}^* = 1$  and  $x_{k_j, v_j}^* = 1$  for some  $v_i, v_j \in V$ ), check if a single vehicle configuration can feasibly serve them together based on the load capacity and route duration constraints, and, if feasible, update the set of clusters by adding  $k_i \cup k_j$ , i.e.,  $K = K \cup \{k_i \cup k_j\}$ . After finishing updating the set of clusters, we solve Problem (3) and repeat this procedure until no further improvement can be reached. Note that, unlike the merging procedure performed during the clustering phase (see Section 4.2), which considers only small clusters that are geographically close, our improvement phase evaluates all possible pairs of clusters (selected in the optimal solution at hand) regardless of their size or spatial proximity.

### 5. FLEETMIX: A Spec-Driven Python Package

FLEETMIX is the executable companion to our matheuristic: a Python package that implements all algorithms and models presented in Section 4. Available on PyPI and at <https://github.com/ekohan/fleetmix> under the MIT license, FLEETMIX serves two audiences. Researchers can reproduce our results with a single command and extend the algorithms by swapping components, and practitioners can deploy our matheuristic directly in production. The version used for this paper (`paper-1.0.0`) is permanently archived to ensure reproducibility.

#### 5.1. Architecture and Design

FLEETMIX implements our matheuristic through four core modules that mirror the paper’s structure: (i) a clustering engine that generates feasible customer clusters using the techniques described in Section 4.2, (ii) an optimization core that solves the integer programming formulation presented in Section 4.3, (iii) a post-optimization refinement module implementing the improvement phase detailed in Section 4.4, and (iv) a benchmarking framework for systematic evaluation.

The implementation uses well-defined interfaces between components. Researchers can substitute alternative clustering algorithms, route time estimators, or MILP solvers without modifying other modules. This modularity is formally defined and enforced by our specification system, which serves as the blueprint for the entire architecture.

#### 5.2. Specification System

To address the semantic gap between mathematical description and software implementation in computational research, we connect the paper’s mathematical formulation to its software through nine machine-readable specification documents. Each specification details a component’s mathematical basis, interface contract, implementation, and usage examples. For example, `clustering.md` documents the split-and-merge procedure (Algorithm 1), its default parameters, and how edge cases

are handled, providing context that is difficult to extract from source code alone. This specification layer enables reviewers to trace equations to implementations and researchers to extend the methodology with confidence.

Table 1 illustrates this traceability, mapping key elements of our matheuristic from the paper to the corresponding software counterparts.

Table 1: Traceability between paper sections, specification documents, and implementation modules.

| Paper Section        | Specification Document                    | Implementation Module                         |
|----------------------|-------------------------------------------|-----------------------------------------------|
| §4.1 Vehicle Configs | <a href="#">vehicle_configurations.md</a> | <code>utils/vehicle_configurations.py</code>  |
| §4.2 Clustering      | <a href="#">clustering.md</a>             | <code>clustering/generator.py</code>          |
| §4.3 Optimization    | <a href="#">optimization.md</a>           | <code>optimization/core.py</code>             |
| §4.4 Improvement     | <a href="#">post_optimization.md</a>      | <code>post_optimization/merge_phase.py</code> |

### 5.3. Usage

FLEETMIX exposes three interfaces. The command-line tool serves as the primary entry point (e.g., `fleetmix optimize --demand customers.csv --config fleet.yaml`), and the command `fleetmix benchmark` regenerates every experiment in Sections 6 and 7. The Python API (`import fleetmix as fm`) enables programmatic integration. The web interface allows interactive parameter exploration without coding. All three interfaces use identical input-output schemas, ensuring reproducibility across modalities. The repository includes worked examples: `custom_clustering.py` implements a round-robin clustering method, `custom_route_time.py` provides a straight-line route time estimator, `custom_solver_adapter.py` integrates a solver with relaxed optimality gap, and `heterogeneous_fleet.py` demonstrates configuring mixed fleets with product-specific vehicle constraints.

## 6. Effectiveness of the Matheuristic Approach

In this section, we present a detailed computational study to test the effectiveness of our matheuristic. We run all experiments on a standard workstation (Apple Silicon M1, 2020) with 8 CPU cores and 16 GB of RAM. The implementation was developed in Python, using Gurobi 12.0.2 as a MILP solver. **Throughout this section, and based on preliminary experiments, we set  $\eta =$  and  $\Delta =$  to define neighboring clusters when applying our merging procedure described in Section 4.2.**

### 6.1. Test Instances

As discussed in Section 2, our work is the first to address a tactical fleet size and mix problem involving heterogeneous MCVs with flexible compartments. Therefore, to the best of our knowledge, there are no benchmark instances or solution approaches in the existing literature. To address this

limitation, we use the publicly available instances generated by [Henke et al. \(2015\)](#) and [Henke et al. \(2019\)](#) for their homogeneous MCVRP with flexible compartments, which are one of the works closest to ours. These instances can be categorized as  $|N|\_|P|\_|M|\_s$ , where  $|N|$  denotes the number of customers,  $|P|$  the number of product types,  $|M|$  the number of compartments, and  $s$  a supply parameter (according to [Henke et al. \(2015\)](#), e.g.,  $s = 1$  means that each customer demands at least one product type and at most one third of the available product types). We consider instances with three product types, three compartments (i.e., each compartment is dedicated to one product type), and  $s \in \{1, 2, 3\}$ . For the instances of [Henke et al. \(2015\)](#), this results in a total of 150 instances with 10 customers and three instances with 50 customers. From [Henke et al. \(2019\)](#), a total of 45 instances are available, five for each number of customers  $|N| \in \{10, 15, \dots, 45, 50\}$ .

To evaluate the effectiveness of our matheuristic, we compare our results with those of [Henke et al. \(2015\)](#) and [Henke et al. \(2019\)](#), which are also publicly available in their manuscript. However, to ensure a fair comparison, we adopt the following considerations, recognizing that we address a tactical fleet size and mix problem with heterogeneous vehicles rather than an operational routing problem with homogeneous vehicles. First, since the cost structure of a fleet size and mix problem is typically driven by fixed vehicle costs, which are not considered by [Henke et al. \(2015\)](#) and [Henke et al. \(2019\)](#), we focus on comparing the number of vehicles to avoid misleading conclusions from incomparable cost structures. Note that the number of vehicles is an input parameter for the MILP model and solution approaches proposed by [Henke et al. \(2015\)](#) and [Henke et al. \(2019\)](#), which they computed by solving a bin packing problem when generating their instances. Second, we only consider homogeneous MCVs (i.e., a single vehicle type) when generating vehicle configurations (see Section 4.1). Third, since [Henke et al. \(2015\)](#) and [Henke et al. \(2019\)](#) do not impose route duration constraints, we do not enforce these constraints when generating feasible customer clusters (see Section 4.2). Lastly, in line with [Henke et al. \(2015\)](#) and [Henke et al. \(2019\)](#), we consider a multi-stop delivery policy when solving Problem (3) (see Section 4.3).

## 6.2. Results

Table 2 summarizes the comparison between the number of vehicles obtained by our matheuristic and those achieved by [Henke et al. \(2015\)](#) and [Henke et al. \(2019\)](#) through their bin packing approach (we refer the reader to our GitHub repository for detailed results). The columns in the table report the total instances per category, the number of instances where our matheuristic achieved the same number of vehicles as [Henke et al. \(2015\)](#) and [Henke et al. \(2019\)](#), the number where our matheuristic found fewer vehicles, and the average run time (in seconds) per instance category.

Table 2 shows the effectiveness of our matheuristic in reducing or maintaining the number of vehicles needed compared to the number of vehicles found by [Henke et al. \(2015\)](#). For instances with 10 customers, our matheuristic finds solutions with exactly one fewer vehicle in 52%, 44%, and 30% of the instances with supply parameters  $s$  equals to 1, 2, and 3, respectively. For larger instances with 50 customers, our matheuristic finds the same number of vehicles as [Henke et al. \(2015\)](#). When compared with [Henke et al. \(2019\)](#), Table 2 reports that our matheuristic consistently finds the same



Table 2: Comparison between the number of vehicles obtained by our matheuristic and those of Henke et al. (2015) and Henke et al. (2019).

| Benchmark           | Category | Instances | # Same vehicles | # Fewer vehicles | Run time (s) |
|---------------------|----------|-----------|-----------------|------------------|--------------|
| Henke et al. (2015) | 10_3_3_1 | 50        | 24              | 26               | 3.9          |
|                     | 10_3_3_2 | 50        | 28              | 22               | 3.8          |
|                     | 10_3_3_3 | 50        | 35              | 15               | 3.9          |
|                     | 50_3_3_1 | 1         | 1               | 0                | 5.1          |
|                     | 50_3_3_2 | 1         | 1               | 0                | 6.1          |
|                     | 50_3_3_3 | 1         | 1               | 0                | 6.8          |
| Henke et al. (2019) | 10_3_3_3 | 5         | 5               | 0                | 3.7          |
|                     | 15_3_3_3 | 5         | 5               | 0                | 3.9          |
|                     | 20_3_3_3 | 5         | 5               | 0                | 3.9          |
|                     | 25_3_3_3 | 5         | 5               | 0                | 4.5          |
|                     | 30_3_3_3 | 5         | 5               | 0                | 3.9          |
|                     | 35_3_3_3 | 5         | 5               | 0                | 4.3          |
|                     | 40_3_3_3 | 5         | 5               | 0                | 4.3          |
|                     | 45_3_3_3 | 5         | 5               | 0                | 4.4          |
|                     | 50_3_3_3 | 5         | 5               | 0                | 5.0          |

number of vehicles, regardless of the problem size. Note that, regardless of the instance category, our matheuristic (from generating vehicle configurations to returning the best solution found; see Figure 1) takes less than eight seconds on average.

In summary, these results validate the effectiveness of our matheuristic in finding high-quality solutions while maintaining short computational times.

## 7. Real-World Case Study

After showing the computational performance of our matheuristic approach, we now present and discuss results from a real-world case study based on the current operations of our industry partner, a leading food distributor based in Bogotá, Colombia. We begin this section by describing our real-world problem instances. We then present and discuss managerial insights regarding the benefits of adopting MCVs. Lastly, we analyze the resulting fleet composition under different cost structures.

Throughout this section, we employ the same parameters and computational environment described in Section 6. **For transparency and reproducibly, our instances and detailed computational results are publicly available at <https://github.com/ekohan/fleetmix>.**

### 7.1. Real-World Problem Instances

Our industry partner provided us with a rich dataset containing 70 days of real-world demand information. Each day (i.e., problem instance) provides detailed information about customer locations and the quantities of products requested by product type (frozen, refrigerated, and dry). Over the 70 days, the number of customers served by the company ranges from 208 to 691, with an average of 379 customers. Below, we explain the main characteristics of the service area and vehicle fleet, as well as the parameter values used.

*Service area.* The service area is located in Bogotá, Colombia, the country's most populous city. Geographically, the service area spans around 1,226 km<sup>2</sup>. The company serves an average of 379 customers per day from a single depot located on the outskirts of the city. Figure 2 shows the spatial distribution of customers for a representative day (to protect confidentiality, the node locations have been slightly perturbed).

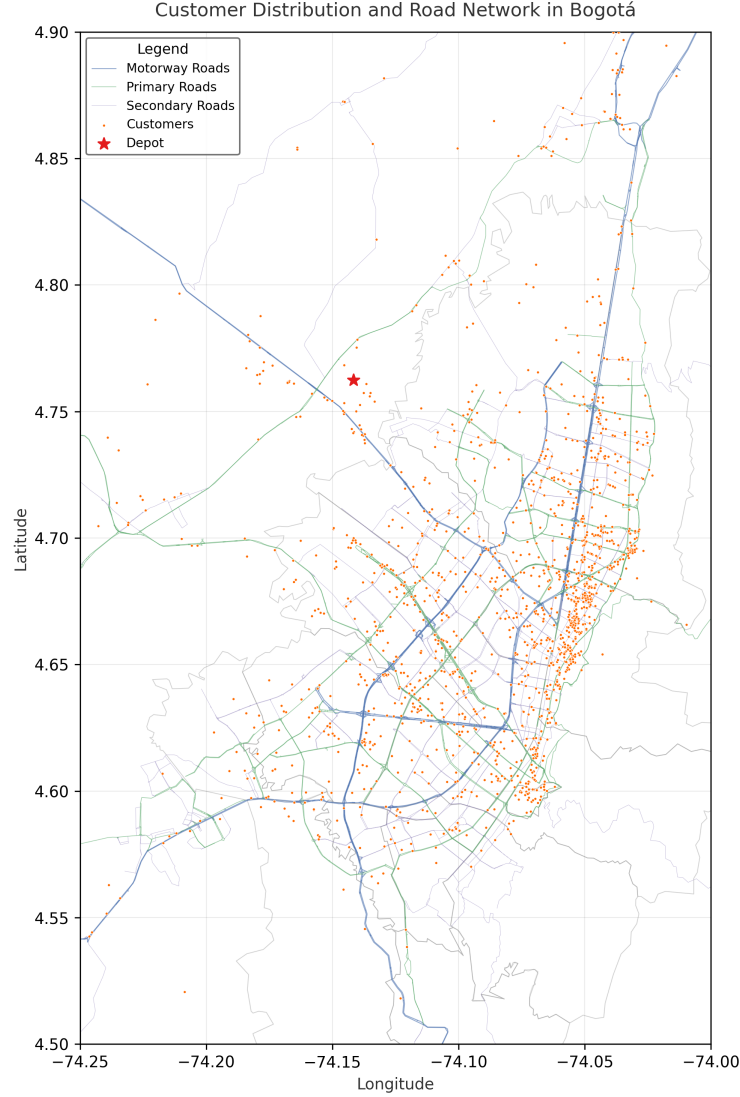


Figure 2: Service area and customer distribution.

*Heterogeneous MCVs.* The company operates a fleet of heterogeneous MCVs, comprising three vehicle types with flexible compartments. Before the delivery horizon begins, the company decides the number of each vehicle type to deploy and their respective compartment configurations. Vehicle type A has a total capacity of 700 kg and a fixed cost of 80 units; type B has a capacity of 1,300 kg and a fixed cost of 140 units; and type C provides 2,500 kg at a fixed cost of 180 units (costs have been scaled for confidentiality). Each vehicle type allows seven compartment configurations

(see Section 4.1), with a (homogeneous) setup cost of 50 units per activated compartment. We assume that vehicles travel the Euclidean distance between nodes at a constant speed of 30 km/h. Based on the company’s recommendation, we set a service time of 25 minutes per stop, which is the average time the company currently takes to deliver food products to their customers (e.g., local restaurants or hotels). The variable travel cost is 10 units per operating hour, with a maximum route duration of 10 hours. **Lastly, given that our instances involve hundreds of customers, we check route duration feasibility through CA when generating customer clusters, using a BHH constant of  $\beta =$  (see Section 4.2.1).**

**JC:** Edited until here.

*Scenarios of analysis.* We set the aforementioned parameters as our baseline scenario, capturing the company’s current operations. However, we also assess the effects of three key parameters on the benefits of adopting MCVs: total vehicle capacity, customer service time, and the maximum allowed vehicle route duration.

**Did we test the impact of employing a single-stop policy?**

## 7.2. Benefits of Using Multi-Compartment Vehicles

The sensitivity analysis assesses how changes in specific parameters influence the fleet size and composition. The analysis focuses on three key parameters: (i) service time, which is the duration it takes for a vehicle to serve a customer at a specific location; (ii) the capacity of the vehicles; (iii) the maximum duration allowed for a route. We modified the parameters in the 70 base instances and compared the key performance indicators (KPIs) by calculating their means to assess the impact throughout each day of the case study. Our initial analysis focuses on how the three parameters influence the number of vehicles selected and the total cost of the operation.

The analysis presented in figure 3 summarizes the results obtained by varying vehicle capacity, service time, and maximum route duration across the 70 base instances. Each parameter was adjusted by a variation of 20% and 50% from the baseline to assess the impact on vehicle utilization and total cost. Figure 3 shows a comparison of these effects in a mixed fleet, with the possibility of being composed of MCVs and SCVs, versus a SCV fleet. The comparison between the two fleet types illustrates the effects of permitting the use of MCVs. In the following paragraphs, we describe how changing the three key parameters affects these fleets.

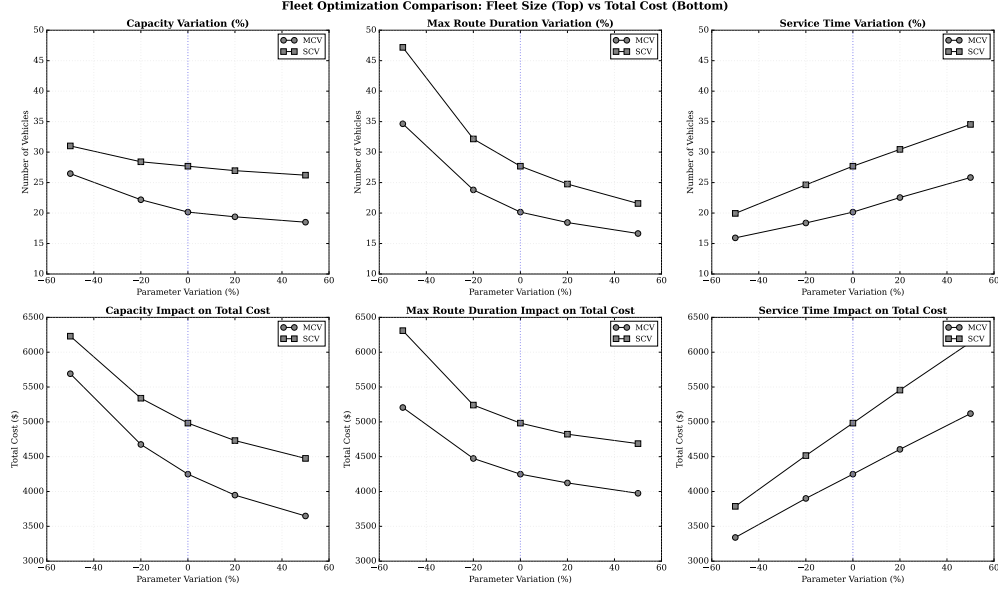


Figure 3: Fleet Size and Total Cost

Effects of Vehicle Capacity. The analysis presented in Figure 3 reveals that increasing vehicle capacity leads to a reduction in the number of vehicles needed for both fleets. In the baseline scenario, the average number of MCVs is 20.2. This number decreases to 19.4 with a 20% increase in capacity and to 18.5 with a 50% increase. For SCVs, the baseline average is 27.7; this decreases to 27 with a 20% capacity increase and to 26.2 with a 50% increase. Conversely, reducing capacity by 20% and 50% results in an increase in fleet size: the number of MCVs rises to 22.2 and 26.5, while the fleet size of SCVs increases to 28.4 and 31, respectively.

The total cost follows a similar trend, decreasing by approximately 8% and 16% under the +20% and +50% scenarios for MCVs, and by 5% and 11% respectively for SCVs. In contrast, the total cost for MCVs increases by 9% and 15% under the -20% and -50% scenarios. Similarly, for SCVs, the total cost also rises as vehicle capacity is reduced; with a 20% reduction, the total cost for MCVs and SCVs increases by 9% and 7%, respectively, while a 50% reduction results in an increase of 25% and 20%, respectively.

Effects of Service Time. The results presented in Figure 3 show that longer service times lead to an increase in the total number of vehicles required in both fleets. For MCVs, the baseline average fleet size increases from 20.2 to 21.6 and 23.8 under +20% and +50% service time scenarios, respectively. Conversely, reducing service time by 20% and 50% decreases the fleet size to 19.2 and 17.8. A similar pattern is observed for SCVs, where the average number of vehicles rises from 27.7 to 29.5 and 32.4 under the +20% and +50% scenarios, and decreases to 26.4 and 24.8 when service time is reduced by 20% and 50%. These results confirm that service duration directly affects fleet size since longer customer interactions reduce the number of customers that can be visited within the same route duration constraint.

Regarding total cost, the effect of service time follows the same trend. The total cost for

MCVs increases by approximately 7% and 15% with +20% and +50% longer service times, while reductions of 20% and 50% lead to cost decreases of around 6% and 13%. For SCVs, the increase is more pronounced, reaching up to 9% and 18% under the +20% and +50,% scenarios. This pattern indicates that MCVs are slightly more efficient at absorbing additional service time due to their capacity to handle multiple products per stop, reducing the need for additional trips compared with single-compartment vehicles.

**Effects of Maximum Route Duration.** Figure 3 also illustrates that increasing the maximum allowable route duration reduces the total number of vehicles required in both fleets. When route duration is extended by 20% and 50%, the fleet size of MCVs decreases from 20.2 to 19.1 and 17.9, respectively. For SCVs, the corresponding values decline from 27.7 to 26.5 and 24.7. Conversely, when the route duration is reduced by 20% and 50%, the number of vehicles rises to 21.8 and 25.6 for MCVs, and to 29.1 and 33.4 for SCVs. This behavior shows that route duration is a critical determinant of fleet requirements, as shorter operational windows constrain route feasibility and require deploying more vehicles to meet demand.

The total cost behavior aligns with these fleet size variations. For MCVs, total cost decreases by roughly 6% and 13% under +20% and +50% duration scenarios, while reductions of 20% and 50% in route duration raise total cost by 8% and 17%. Similarly, for SCVs, total cost decreases by 5% and 11% with longer durations and increases by 9% and 19% when duration is reduced. These results demonstrate that extending route duration enables better vehicle utilization and reduces total operating costs, particularly benefiting the heterogeneous fleet configuration of MCVs.

Table 3: Average operational metrics (over 70 instances) per parameter variation.

| Parameter           | Variation | Load vehicle utilization |       | Customers per vehicle |      | Effective route duration |     |
|---------------------|-----------|--------------------------|-------|-----------------------|------|--------------------------|-----|
|                     |           | MCV                      | SCV   | MCV                   | SCV  | MCV                      | SCV |
| Capacity            | -50%      | 87.0%                    | 85.9% | 10.0                  | 12.2 | 5.3                      | 6.6 |
|                     | -20%      | 83.6%                    | 79.6% | 11.7                  | 13.3 | 6.1                      | 7.1 |
|                     | 0%        | 81.4%                    | 76.0% | 12.8                  | 13.7 | 6.6                      | 7.3 |
|                     | 20%       | 79.7%                    | 71.8% | 13.3                  | 14.1 | 6.9                      | 7.5 |
|                     | 50%       | 76.1%                    | 66.5% | 13.8                  | 14.4 | 7.1                      | 7.7 |
| Service Time        | -50%      | 83.6%                    | 78.4% | 16.9                  | 19.1 | 5.0                      | 5.9 |
|                     | -20%      | 82.6%                    | 76.9% | 14.2                  | 15.4 | 6.1                      | 6.8 |
|                     | 0%        | 81.4%                    | 76.0% | 12.8                  | 13.7 | 6.6                      | 7.3 |
|                     | 20%       | 80.4%                    | 73.9% | 11.4                  | 12.4 | 6.9                      | 7.7 |
|                     | 50%       | 78.5%                    | 70.3% | 10.0                  | 10.9 | 7.3                      | 8.1 |
| Max. Route Duration | -50%      | 70.6%                    | 57.1% | 7.4                   | 8.0  | 4.0                      | 4.4 |
|                     | -20%      | 80.4%                    | 72.3% | 10.9                  | 11.8 | 5.7                      | 6.3 |
|                     | 0%        | 81.4%                    | 76.0% | 12.8                  | 13.7 | 6.6                      | 7.3 |
|                     | 20%       | 82.5%                    | 78.2% | 14.1                  | 15.3 | 7.3                      | 8.1 |
|                     | 50%       | 82.3%                    | 78.9% | 15.7                  | 17.6 | 8.0                      | 9.3 |

The previous analysis examined how variations in capacity, service time, and route duration

affect total cost and fleet size. To enhance our findings, Table 3 includes additional indicators such as load vehicle utilization, the number of customers served per vehicle, and effective route duration. These additional indicators complement the analysis of the parameters' impact on the total cost and fleet size.

Load vehicle utilization decreases as vehicle capacity and route duration increase. For multi-compartment vehicles (MCVs), utilization drops from 87% to 76.1% as capacity increases from -50% to +50%. Similarly, for single-compartment vehicles (SCVs), utilization declines from 85.9% to 66.5%. A comparable trend occurs with increasing route duration; utilization is higher under restrictive conditions. In contrast, longer service times lead to a reduction in utilization, with figures decreasing from 83.6% to 78.5% for MCVs and from 78.4% to 70.3% for SCVs. These results indicate that utilization adjusts based on the operational constraints of each parameter, which aligns with the observed changes in total cost and fleet size. **VS: double check this paragraph**

The number of customers per vehicle and the effective route duration vary consistently with the analyzed parameters. Reducing service time increases the number of customers served per vehicle, while longer service times produce the opposite effect. When the maximum route duration increases, both indicators rise, showing longer operational routes and a higher number of customers visited. Conversely, reductions in route duration lead to shorter routes and fewer customers per vehicle. For all scenarios, SCVs serve a larger number of customers per route, while MCVs operate with slightly shorter effective durations. The combined behavior of these indicators corresponds to the cost and fleet trends presented in the previous figures.

Overall, the results from Table 3 and Figure 3 demonstrate that operational parameters—capacity, service time, and route duration—directly influence the fleet mix and utilization level. The next sensitivity analysis extends this exploration to economic parameters, specifically the fixed cost of multi-compartment vehicles and the compartment setup cost. These factors determine not only the total fleet cost but also the cost effectiveness of using MCV over SCV, shaping the overall composition and mix of the fleet.

### 7.3. Fleet Composition Analysis

This subsection analyzes how compartment-related costs affect fleet composition, specifically the share of (MCVs) within the total fleet. The analysis varies two cost parameters: the fixed cost associated with selecting an MCV and the compartment setup cost. These parameters are increased progressively to observe how the proportion of MCVs changes across the 70 evaluated instances, each representing one day of operation. The results are presented through two figures that illustrate the impact of these cost variations on the use of MCVs, which are discussed in the following paragraphs.

The results are summarized in Figure 4, which presents the combined effect of the compartment setup cost and the multiplier on the share of multi-compartment vehicles (MCVs) in the fleet ( $\alpha$ ). The horizontal axis represents the compartment setup cost, which increases from 0% to 50% in

increments of 10%. The vertical axis corresponds to the  $\alpha$  multiplier, which scales the fixed cost of selecting an MCV from 1.0 to 2.0 in increments of 0.1. Each cell in the heat map displays three values: the average MCV share in percentage, the number of days (out of 70) in which at least one MCV was used, and the number of days in which all vehicles in the fleet were MCVs, marked with a star.

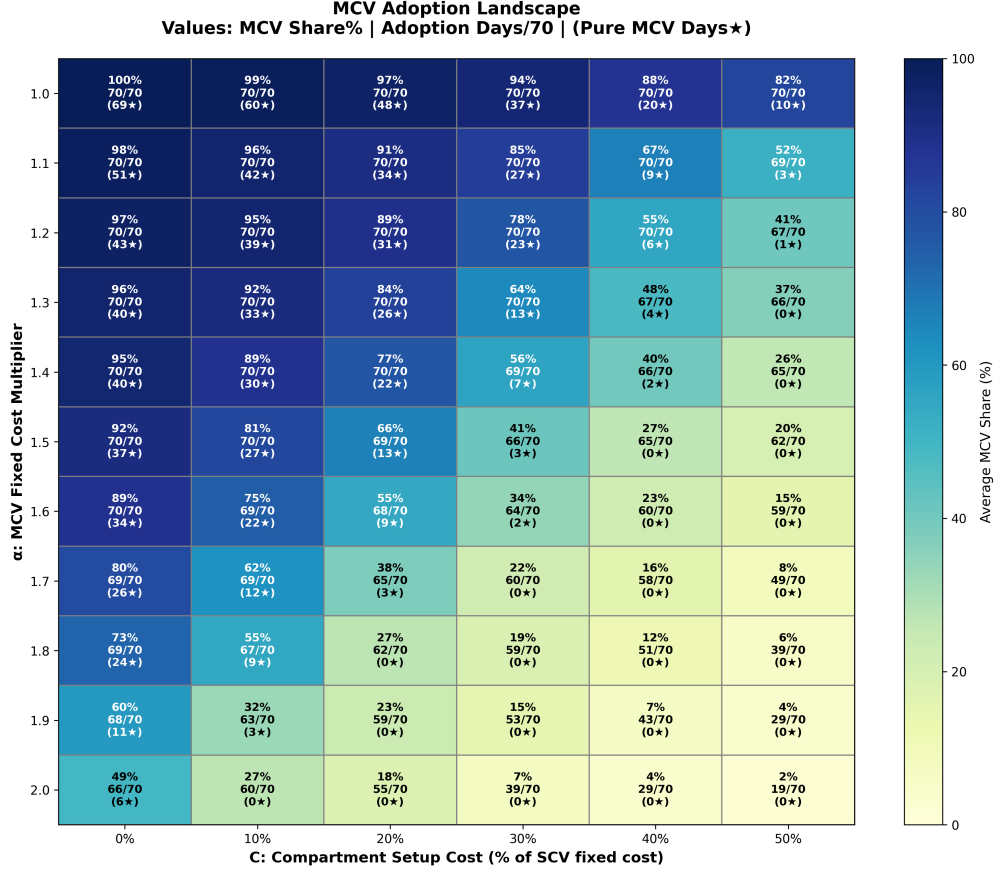


Figure 4: MCVs Adoption Landscape

The figure shows that as both  $\alpha$  and the compartment setup cost increase, the average MCV share in the fleet decreases progressively. For low cost values ( $\alpha = 1.0$  and  $C = 0\%$ ), the fleet is entirely composed of MCVs across all instances. As costs rise, the share of MCVs declines continuously, with a larger number of mixed or single-compartment fleet configurations appearing. When  $\alpha$  reaches 2.0 and  $C = 50\%$ , the average share of MCVs falls below 10%, and only a few instances include any MCV. The distribution of values across the grid indicates that higher costs lead to a consistent reduction in the proportion of multi-compartment vehicles in the fleet.

Figure 5 provides a detailed view of how the MCV share varies across the 70 daily instances under different compartment setup cost scenarios. The horizontal axis represents the day index, and the vertical axis indicates the proportion of MCVs in the fleet for each day. All three plots correspond to  $\alpha = 1.6$ , meaning that the fixed cost of each MCV is 1.6 times higher than the baseline. The left plot represents the baseline condition ( $C = 0\%$ ), the middle plot corresponds to

a 20% increase in compartment setup cost, and the right plot shows the 50% increase case.

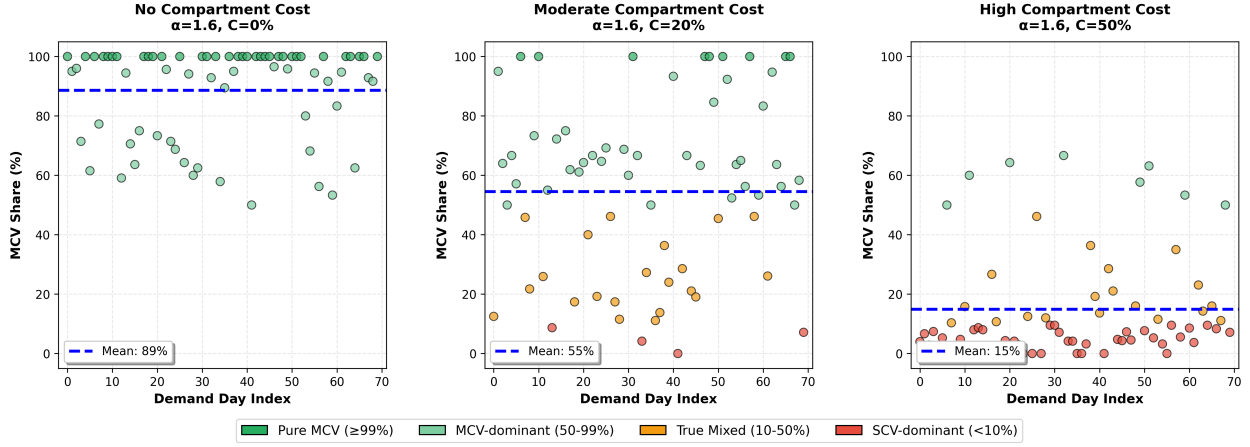


Figure 5: MCVs Share in the Fleet

In the baseline scenario, most days exhibit a high MCV share, with the average reaching approximately 89%. When the compartment setup cost increases to 20%, the daily MCV share becomes more heterogeneous, with values ranging between 20% and 100% and an overall mean of 55%. Under the 50% setup cost scenario, the MCV share drops considerably, with most of the days showing less than 10% MCV usage and a mean value close to 15%. These results indicate that as the compartment setup cost increases, the presence of MCVs in the fleet progressively diminishes across daily operations, consistent with the overall pattern observed in Figure 4.

The analysis of fleet composition demonstrates the sensitivity of multi-compartment vehicle (MCV) adoption to variations in compartment setup and MCVs selection fixed costs. The following section, Discussion, examines the managerial implications of these results, focusing on the operational benefits and considerations of incorporating MCVs in food distribution fleets operating within urban environments.

## 8. Discussion

- MCV

Summarize the main managerial implications of the work.

## 9. Conclusions

Multi-Compartment Vehicles (MCVs) are among the most effective vehicle technologies that food distributors can employ to consolidate deliveries of multiple product types while maintaining their specific temperature and loading conditions. Motivated by our collaboration with a major food distributor operating in Bogotá, Colombia, we investigate the design of last-mile delivery fleets



involving heterogeneous MCVs with flexible compartment sizes. Specifically, the decision-making problem faced by the company considers three interconnected decisions: (i) the number of each vehicle type to use, (ii) the type and size of each compartment assigned to each vehicle, and (iii) the distribution plan for each vehicle. The objective is to minimize total logistics costs, which comprise both fixed and variable costs associated with selecting and utilizing vehicles.

To solve this problem, we propose a novel matheuristic approach based on a cluster-first, fleet design-second scheme. Our matheuristic can accommodate several real-world characteristics of modern food distribution operations, including homogeneous and heterogeneous vehicles with fixed or flexible compartment sizes; a comprehensive cost structure that allows including fixed costs, compartment-dependent costs, and variable (distance or time-dependent) costs; and single- and multiple-stop delivery policies. To facilitate the application of our matheuristic, we create FLEETMIX, an open-source Python package that is publicly available under the MIT license and distributed via PyPI. **FLEETMIX follows state-of-the-practice software engineering practices and features a modular-based architecture that allows replacing individual components while maintaining the integrity of the overall pipeline.**

Using established synthetic instances from the literature, we show that FLEETMIX achieves high-quality fleet designs compared to existing solution approaches. Through a real-world case study informed by real data from our partner industry company, we derive several managerial insights into the value of using MCVs to reduce total logistics costs and the number of vehicles needed to deliver multiple product types. We show that **[summarize main insights.]**

Since our work is among the first to address the tactical design of last-mile delivery fleets with MCVs, there are several opportunities for future work. First, while this is a challenging optimization problem, future research could focus on developing exact solution approaches that can solve practical problem instances to proven optimality. Here, one could develop a column generation-based scheme and use our set covering formulation presented in Section 4. Second, future work can also extend FLEETMIX to accommodate other real-world characteristics, such as other cost components (e.g., loading and unloading costs, as in [Hübner and Ostermeier \(2019\)](#)) and time windows. Lastly, a stochastic version of the problem, where tactical fleet size and mix decisions are made before customer demand is known, will be a challenge to address.

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## Appendix A. Clustering Techniques

To create customer clusters, we use a variety of methods, including  $k$ -means,  $k$ -medoids, Gaussian mixture, and hierarchical clustering (agglomerative). These methods can be used individually or in combination through what we call the *combine* method. Table A.4 summarizes and compares these clustering techniques.

Table A.4: Comparison of Clustering Methods

| Method           | Weight                             | Parameters                            | Advantages                                                       |
|------------------|------------------------------------|---------------------------------------|------------------------------------------------------------------|
| $k$ -means       | $\lambda = 1$                      | Number of clusters                    | Fast convergence, simple to implement                            |
| $k$ -medoids     | $\lambda = 1$                      | Number of clusters, medoid selection  | Robust to outliers                                               |
| Agglomerative    | $\lambda \in \{0, 0.2, \dots, 1\}$ | Linkage criteria and $\lambda$ weight | Captures both spatial and demand factors                         |
| Gaussian Mixture | $\lambda = 1$                      | Number of clusters, covariance type   | Flexible cluster shapes, probabilistic cluster assignment        |
| Combine          | $\lambda \in \{0, 0.2, \dots, 1\}$ | All above parameters                  | Ensemble approach, robust solutions, diverse clustering patterns |